Hospital Patient Flow, Congestion & Bed Capacity Planning

Applied Stochastic Models (ASM) Course Project — Models, Datasets, Algorithms & Deep Dives

Course: Applied Stochastic Models (Fall 2026)

Primary Datasets: NYS SPARCS (250k discharges) + PhysioNet MIMIC-IV Demo v2.2

Repository: github.com/Manas-Jain-24/applied-stochastic-models-hospital-flow

Verification: 38 Automated Unit Tests (100% Passing) + Replicated Priority DES

Executive Master Matrix: Questions, Models, Datasets & Findings

Module / Q#	Topic & Scope	Stochastic Model	Dataset Used	Algorithm / Method	Key Observation / Numerical Finding
Q1	Memoryless Test & Heavy Tails	$M/G/c$ Heavy-Traffic & LOS distributions	NYS SPARCS (250k records)	MLE fitting, K-S test, AIC/BIC, Allen–Cunneen	$c_v^2 = 1.991 \approx 2.0$. Lognormal/Gamma best. Variance creates a **50% delay penalty** over $M/M/c$.
Q2	Inpatient Bed Dimensioning	$M/M/c/c$ Erlang Loss System (CTMC Birth-Death)	NYS SPARCS ($a=269.3$ bed-days)	Downward stable Erlang-B recursion	Nominal $c=269$ causes 4.78% blocking (~872 lost/yr). Target $P_{\text{block}}\le 1\%$ requires **$c^* = 293$ beds** (+24 buffer).
Q3	Emergency vs. Elective Triage	Non-preemptive Multi-Server Priority Queue	SPARCS split & MIMIC-IV ED triage	Cobham priority heuristic via Erlang-C delay	Cuts emergency wait ~3x (1.47h vs 12.33h elective; or 6.0 min vs 74.4 min at $c=293$) under work conservation.
Q4	Diurnal Arrival Clustering	NHPP with 24h harmonic intensity $\lambda(t)$	SPARCS census & MIMIC-ED timestamps	IDC dispersion test & Lewis–Shedler Thinning	Overdispersed ($IDC=1.34>1$). Peaks at **2:12 PM** (3.25/hr) vs **2:12 AM** (0.91/hr) — a **3.55x surge**.
Q5	Finite ED Holding & Diversion	$M/M/c/K$ Finite Queue ($K = c + B$)	NYS SPARCS ($a=269.3$, $c=285$)	CTMC steady-state with Log-Sum-Exp (LSE)	0 buffer $\to$ 12.4% diversion. Adding 16 beds drops diversion by **77%** (to 2.8%) at 3.6h boarding wait cost.
Q6	Downstream Patient Transitions	Absorbing Markov Chain ($P=[Q\ R; 0\ I]$)	SPARCS dispositions & MIMIC transfers	Fundamental matrix $N=(I-Q)^{-1}$, $B=NR$, MFPT	Direct ICU admits have a **10.1% mortality** and **33.2% SNF transfer** probability; 1.48 total ward transfers.
Q7	Epidemic Surge & Staff Shortage	Multi-server Stability ($\rho < 1.0$) & DES	SPARCS baseline under synthetic shocks	Boundary testing & event-calendar simulation	Baseline $\rho=92.9\%$. +20% surge pushes $\rho=111.4\%$; -10% staff pushes $\rho=103.2\%$. System collapses; requires throttling electives.
Q8	Multi-Stage Department Flow	4-Station Open Jackson Queueing Network	SPARCS routing & MIMIC `transfers.csv`	Traffic equations $(I - P^T)\vec{\Lambda} = \vec{\gamma}$, Product Form	**Medical/Surgical** is hospital bottleneck ($\rho = 92.2\%$). Network-wide end-to-end stay is **5.19 days**.
Q9	Diurnal Dynamic Shift Staffing	Pointwise Stationary Approximation (PSA)	Calibrated diurnal profile $\lambda(t)$	Hourly Erlang-C & 3-shift dynamic matching	Constant staffing causes 41.5 min peak wait; shifting 25 beds from night to day cuts wait to **1.1 min (-97.3%)** at **$0 cost**.
Q10	Multi-Ward Bed Allocation	Integer Convex Resource Allocation (Erlang-B)	SPARCS & MIMIC-IV multi-ward loads	**Fox's Greedy Marginal Allocation Algorithm**	Proportional allocation starves ICU (6.66% loss). Fox allocation saves **83.8%–85.1% weighted blocking** via $\sqrt{a}$ buffering.
Q11	MIMIC Pipeline Integration	Empirical Stochastic Calibration Pipeline	PhysioNet MIMIC-IV & ED Demo v2.2	Timestamp dwell extraction, ETL, JSON export	ICU LOS ($E[S]=3.68$d, $c_v^2=1.12$), ESI acuity (8.7% ESI-1), 5.45% mortality. 100% drop-in scalable to full MIMIC-IV.
DD 1	Heavy-Tail & Hazard Kinetics	Hazard rate $h(t)$, P–K $M/G/1$, $M/G/c$	NYS SPARCS ($N=250\text{k}$, $c_s^2=1.9915$)	MLE log-likelihood, large-sample K-S, $\Phi(c_s^2)$	Lognormal hazard matches clinical convalescence. Multiplier $\Phi(c_s^2)=1.496$ accounts for the ~49.6% delay penalty.
DD 2	Capacity, Priority & DES Audit	Erlang-B, Priority Heuristic, Priority DES	NYS SPARCS & MIMIC ED triage	Jagerman recursion & 10-replication priority DES	DES confirms emergency wait 5.29 ± 1.56 min vs 57.41 ± 35.10 min elective (0 censored pts). Halfin–Whitt $\beta=1.44$.
DD 3	Diurnal Shift & Network Stability	NHPP, PSA Shifts, Jackson Net, Fox Optimizer	SPARCS & MIMIC arrival/transfer flows	Lewis–Shedler thinning, linear solver, Fox greedy	Exact $\vec{\Lambda}=[40.0, 6.23, 8.54, 30.13]$. Nominal MedSurg ($c=120$) is unstable ($\rho_4=125.5\%$); requires ≥ 151 beds.

Detailed Breakdown of the 11 Core Research Questions

Comprehensive technical reference for mathematical formulation, empirical data grounding, computational algorithms, and quantitative findings.

Question 1: The Exponential Service Assumption & Heavy-Tail Penalty

Stochastic Model: Parametric Length of Stay (LOS) Distributions & Allen–Cunneen M/G/c Queueing Model

Dataset Used: NYS SPARCS Inpatient Discharges ($N = 250,000$ patient records; $E[S]=5.386$ days, $\text{Var}(S)=57.77$ days^2, $\sigma=7.60$ days).

Algorithm / Solver: Maximum Likelihood Estimation (MLE) parameter estimation for Exponential, Gamma, Lognormal, and Weibull; Kolmogorov–Smirnov (K-S) goodness-of-fit testing; Akaike & Bayesian Information Criteria (AIC/BIC); Allen–Cunneen variance correction $\Phi(c_v^2) = \frac{1 + c_v^2}{2}$.

Key Observation: Empirical length of stay exhibits $c_v^2 = 1.991 \approx 2.0$. The memoryless property is decisively rejected ($p < 10^{-15}$ for Exponential; Lognormal is best by AIC/BIC). Under the Allen–Cunneen approximation, recovery variance inflates expected queue delays by **50%** ($W_q(M/G/c) \approx 1.50 \times W_q(M/M/c)$), proving that traditional $M/M/c$ models dangerously underestimate hospital congestion.

Implementation: `src/distributions.py & src/queueing.py (mgc_allen_cunneen)`

Question 2: Bed Capacity Sizing via Erlang-B Loss Modeling

Stochastic Model: M/M/c/c Continuous-Time Markov Chain (CTMC) Birth–Death Loss System (Erlang-B)

Dataset Used: NYS SPARCS (calibrated arrival rate $\lambda = 50.0$ admissions/day, mean stay $E[S] = 5.386$ days $\implies$ offered load $a = 269.3$ bed-days / Erlangs).

Algorithm / Solver: Numerically stable downward Erlang-B recurrence: $B(k, a) = \frac{a B(k-1,a)}{k + a B(k-1,a)}$ with $B(0, a) = 1.0$ (avoids factorial float overflow for $c \approx 300$).

Key Observation: Sizing capacity to exactly equal average demand ($c = 269$ beds) causes a 4.78% blocking probability (~872 diverted patients/year). Guaranteeing $P_{\text{block}} \le 1.0\%$ requires **$c^* = 293$ beds** (a $+23.7$-bed safety buffer, bed occupancy = 91.9%). An acute care hospital cannot operate at 100% average occupancy without catastrophic patient turnaway.

Implementation: `src/queueing.py (erlang_b, min_capacity_erlang_b)`

Question 3: Two-Class Non-Preemptive Priority Triage Queueing

Stochastic Model: Multi-Server Non-Preemptive Priority Queueing ($M/M/c$ with Emergency Class 1 vs. Elective Class 2)

Dataset Used: NYS SPARCS admission mix (76.7% emergent/urgent vs. 23.3% elective) & MIMIC-IV ED triage distributions.

Algorithm / Solver: Cobham's multi-server priority queueing heuristic using the Erlang-C delay probability $W_0 = \frac{C(c, a)}{c\mu}$: $W_{q,1} = \frac{W_0}{1 - \rho_1}$ and $W_{q,2} = \frac{W_0}{(1 - \rho_1)(1 - \rho)}$.

Key Observation: Prioritization protects emergency admissions without adding beds. At $c = 293$ beds, emergency wait drops to **6.0 minutes** (0.10 hr) while schedulable elective wait is **74.4 minutes** (1.24 hr), compared to an unprioritized FCFS wait of 33.4 minutes. Weighted work conservation holds: $\frac{\lambda_1 W_{q,1} + \lambda_2 W_{q,2}}{\lambda_1 + \lambda_2} = W_q^{\text{FCFS}}$.

Implementation: `src/queueing.py (priority_queue_two_class)`

Question 4: Non-Homogeneous Poisson Arrivals & Diurnal Thinning

Stochastic Model: Non-Homogeneous Poisson Process (NHPP) with 24-Hour Harmonic Intensity Function $\lambda(t)$

Dataset Used: NYS SPARCS arrival census records & PhysioNet MIMIC-IV-ED encounter timestamps.

Algorithm / Solver: Index of Dispersion for Counts (IDC) overdispersion test; non-linear harmonic least-squares fitting; **Lewis–Shedler Acceptance–Rejection Thinning Algorithm** with majorizing rate $\lambda^* = \max \lambda(t)$.

Key Observation: Empirical arrivals are significantly overdispersed ($IDC = 1.34 > 1.0$, rejecting homogeneous Poisson). Intensity follows $\lambda(t) = 2.08 + 1.17 \cos(\frac{2\pi(t - 14.2)}{24})$, peaking at **2:12 PM** (3.25 arr/hr) and dropping to **2:12 AM** (0.91 arr/hr) — a **3.55x peak-to-trough surge** that drives severe afternoon congestion.

Implementation: `src/arrival_process.py (index_of_dispersion_for_counts, simulate_nhpp_thinning)`

Question 5: Finite-Buffer M/M/c/K ED Holding & Ambulance Diversion

Stochastic Model: M/M/c/K Finite-Capacity Queueing System (c inpatient beds, $K = c + B_{\text{buffer}}$ total hospital limit)

Dataset Used: NYS SPARCS ($a = 269.3$ bed-days, $c = 285$ inpatient beds, variable holding buffer $B \in [0, 20]$).

Algorithm / Solver: CTMC birth–death steady-state solver with **Log-Sum-Exp (LSE)** normalization to prevent numerical underflow/overflow for large state spaces ($K > 300$); Little's Law for effective arrival rate $\lambda_{\text{eff}} = \lambda(1 - p_K)$.

Key Observation: Directly quantifies the clinical trade-off: with 0 holding beds ($K=285$), ambulance diversion is 12.4% with zero boarding wait. Adding 16 holding beds ($K=301$) drops ambulance diversion by **77%** (to 2.8%) at the expense of an average 3.6-hour ED boarding delay.

Implementation: `src/queueing.py (mmck_finite_buffer_metrics)`

Question 6: Absorbing Markov Chains for Downstream Patient Transitions

Stochastic Model: Discrete-Time Absorbing Markov Chain with Canonical Partition $P = [Q\ R; 0\ I]$

Dataset Used: NYS SPARCS ($250,000$ discharge encounter dispositions) & MIMIC-IV care-unit transfer paths.

Algorithm / Solver: Canonical decomposition across 3 transient wards (Med, Surg, ICU) and 4 absorbing states (Home, Home Health, SNF, Expired); Fundamental matrix inversion $N = (I - Q)^{-1}$; Absorption probability matrix $B = NR$; Expected steps to absorption $t = N\mathbf{1}$; Kemeny–Snell Mean First Passage Time (MFPT).

Key Observation: Direct ICU admissions face an in-hospital mortality probability of **10.1%** and a **33.2% probability of requiring a Skilled Nursing Facility (SNF)**. An ICU patient averages 1.16 visits in ICU, 0.47 in Medical, and 0.31 in Surgical (1.48 total ward transfers) before final discharge.

Implementation: `src/markov_chain.py (analyze_absorbing_markov_chain)`

Question 7: Operational Stress-Testing & Epidemic Surge Collapse

Stochastic Model: Multi-server Stability Analysis ($\rho = \frac{\lambda}{c\mu} < 1.0$) and Event-Calendar DES

Dataset Used: Calibrated NYS SPARCS parameters under simulated winter epidemic surge and staffing restriction scenarios.

Algorithm / Solver: Traffic intensity boundary evaluation and fast Discrete-Event Simulation (`heapq` event calendar).

Key Observation: A hospital at baseline $\rho = 92.9\%$ cannot absorb shocks: a **+20% winter demand surge** pushes $\rho = 111.4\%$, and a **-10% staffing shortage** pushes $\rho = 103.2\%$. In both cases, queues explode toward infinity. Operational survival mathematically requires **actively throttling schedulable elective admissions** (which represent 23.3% of volume) when occupancy reaches 95%.

Implementation: `src/simulation.py & src/queueing.py`

Question 8: Multi-Stage Open Jackson Queueing Networks

Stochastic Model: 4-Station Open Jackson Queueing Network (Emergency, ICU, Step-Down, Medical/Surgical)

Dataset Used: Inter-unit transfer matrices derived from NYS SPARCS APR-DRG and MIMIC-IV `transfers.csv.gz`.

Algorithm / Solver: Linear system solver for Jackson traffic equations $(I - P^T)\vec{\Lambda} = \vec{\gamma}$; Jackson's Product-Form Theorem; station-level $M/M/c_j$ performance; Network Little's Law $W_{\text{total}} = \frac{\sum L_j}{\sum \gamma_j}$.

Key Observation: Captures cross-departmental spillover: **Medical/Surgical** is the primary system bottleneck ($\rho = 92.2\%$ in baseline report; $\rho_4 = 125.5\%$ unstable under nominal deep-dive parameters requiring ≥ 151 beds to stabilize). Network-wide end-to-end hospital residence time is **5.19 days**.

Implementation: `src/jackson_network.py (solve_traffic_equations, analyze_jackson_network)`

Question 9: Time-Varying Pointwise Stationary Approximation (PSA) & Shift Staffing

Stochastic Model: Pointwise Stationary Approximation (PSA) for Non-Stationary Queues ($M_t/M/c_t$)

Dataset Used: Calibrated diurnal arrival intensity $\lambda(t)$ from SPARCS / MIMIC-ED.

Algorithm / Solver: Hourly numerical evaluation of instantaneous offered load $a(t) = \lambda(t)/\mu$ and Erlang-C waiting time $W_q(t)$ across 24 hourly slices; Discrete shift optimization matching bed capacity $c(t)$ to demand.

Key Observation: Constant 24/7 staffing (18 beds flat) causes an afternoon wait spike of **41.5 minutes** ($\rho = 94.0\%$). Dynamically shifting 25 beds from quiet night hours to peak daytime hours (14 night / 22 day / 18 evening) under the **exact same total bed-hours (cost-neutral)** cuts peak wait to **1.1 minutes**—a **97.3% delay reduction** at zero extra budget.

Implementation: `src/time_varying.py (compare_constant_vs_dynamic_staffing)`

Question 10: Multi-Department Optimal Bed Allocation via Fox's Greedy Algorithm

Stochastic Model: Separable Integer Convex Optimization under Erlang-B Loss $\min \sum w_j B(c_j, a_j)$ subject to $\sum c_j = C$

Dataset Used: Multi-department offered loads ($a_{\text{ICU}}=20, a_{\text{ED}}=30, a_{\text{MedSurg}}=150$ Erlangs) and clinical severity weights ($w=[4, 2, 1]$).

Algorithm / Solver: Fox's Greedy Marginal Allocation Algorithm (exploiting strict discrete convexity of Erlang-B to guarantee the global integer optimum); marginal gradient tracking $\Delta_j = w_j [B(c_j, a_j) - B(c_j + 1, a_j)]$.

Key Observation: Naive proportional sizing under-buffers the ICU (gives 24 beds $\to$ 6.66% rejection rate). Fox's algorithm honors the square-root staffing principle ($\text{Buffer} \propto \sqrt{a}$), giving ICU 33–40 beds (0.14% rejection) and reducing weighted hospital blocking by **83.8%–85.1%**.

Implementation: `src/bed_allocation.py (marginal_bed_allocation, proportional_bed_allocation)`

Question 11: Modular MIMIC-IV & MIMIC-ED Pipeline Integration

Stochastic Model: Empirical Stochastic Calibration & Automated ETL Data Pipeline

Dataset Used: PhysioNet open-access **MIMIC-IV Clinical Demo v2.2** & **MIMIC-IV-ED Demo v2.2** (`admissions`, `transfers`, `icustays`, `triage`).

Algorithm / Solver: Automated dwell-time extraction from timestamps, care-unit movement trajectory reconstruction, empirical transition matrix generation, and JSON parameter export (`src/mimic_adapter.py`).

Key Observation: Extracts real-world ICU length of stay ($E[S]=3.68$ days, $c_v^2=1.12$), emergency triage mix (ESI-1: 8.7%, ESI-2: 46.9%, ESI-3: 43.5%), and 5.45% mortality. Architecture is **100% drop-in scalable** to full credentialed MIMIC-IV with zero code alterations.

Implementation: `src/mimic_adapter.py (run_mimic_pipeline)`

The Three Specialized Deep-Dive Investigation Tracks

Graduate-level mathematical scrutiny, hazard rate dynamics, continuous-time queueing derivations, and simulation audits.

Deep Dive Track 1: Heavy-Tailed Service Times, Hazard Rate Dynamics, and the M/G/c Delay Penalty

Expands Question 1 & Flagship Research Theme 3

Stochastic Framework: Hazard Rate Kinetics $h(t) = \frac{f(t)}{1 - F(t)}$, Pollaczek–Khinchine $M/G/1$ Exact Variance Expansion, Allen–Cunneen Multi-Server Heavy-Traffic Approximation, and Service Variance Sensitivity Sweeps ($c_s^2 \in [0.1, 3.5]$).

Empirical Datasets: NYS SPARCS Inpatient Discharges ($N = 250,000$ discharge records; $E[S] = 5.386$ days, $\text{Median} = 3.0$ days, $\text{Var}(S) = 57.771$ days^2, $\sigma = 7.601$ days, $c_s^2 = 1.9915$).

Algorithms & Methods: Maximum Likelihood Estimation (MLE) log-likelihood evaluation, AIC/BIC information criteria, large-sample Kolmogorov–Smirnov power audit, and the Pollaczek–Khinchine variance expansion $W_q(M/G/c) \approx W_q(M/M/c) \cdot \frac{1 + c_s^2}{2}$.

Key Discoveries & Audited Findings:
1. **Hazard Rate Mechanics:** Exponential hazard is memoryless ($h(t)=\mu$), failing to capture biological recovery. Lognormal hazard accurately rises from 0 to a peak during modal recovery, then asymptotically declines for complicated outliers.
2. **The Allen–Cunneen Multiplier:** With $c_s^2 = 1.9915$, $\Phi(c_s^2) = \frac{1 + 1.9915}{2} \approx 1.496$. Holding mean stay and bed capacity fixed, real hospital delays are **49.6% higher** than standard $M/M/c$ predictions.
3. **Methodological Audit:** Clarified that with $N=250,000$, formal hypothesis tests reject any parametric family ($p < 10^{-15}$) due to massive statistical power; model selection must rely on penalized information criteria (AIC/BIC), where Lognormal is superior.

Source Document: `deep_dives/01_heavy_tail_service_deep_dive.md`

Deep Dive Track 2: Bed Capacity Sizing, Multi-Class Priority Triage, and Finite-Buffer Holding

Expands Questions 2, 3, and 5 & Flagship Research Theme 2

Stochastic Framework: CTMC Birth–Death $M/M/c/c$ Loss Model, Two-Class Non-Preemptive Multi-Server Priority Queueing, $M/M/c/K$ Finite-Buffer System, and Replicated Event-Calendar Priority DES.

Empirical Datasets: NYS SPARCS ($a = 269.3$ Erlangs, 50 admissions/day, 60% emergency vs. 40% elective) and PhysioNet MIMIC-IV ED triage distributions.

Algorithms & Methods: Jagerman's inverse downward recurrence for Erlang-B; Cobham's multi-server priority heuristic ($W_0 = C(c,a)/c\mu$); **Drained-horizon 10-replication priority discrete-event simulation** (`src/priority_simulation.py`) with 0 right-censored patients; Log-Sum-Exp stabilized $M/M/c/K$ solver.

Key Discoveries & Audited Findings:
1. **Erlang-B Sizing Reconciliation:** Sizing at load ($c=269$) causes 4.78% blocking (~872 diverted/yr). Target $c^* = 293$ achieves 0.91% blocking (~166 diverted/yr) with a $+23.7$-bed ($+8.8\%$) safety buffer.
2. **Halfin–Whitt Regime:** Capacity sizing satisfies the square-root staffing principle $c^* \approx a + \beta \sqrt{a}$ with safety parameter $\beta \approx 1.44$.
3. **Priority DES Validation:** Replicated simulation (730-day arrival horizon + drained observation) yields emergency wait $W_{q,1} = 5.29 \pm 1.56$ min vs. elective wait $W_{q,2} = 57.41 \pm 35.10$ min, confirming analytical heuristic bounds with zero right-censoring bias.

Source Document: `deep_dives/02_capacity_and_priority_deep_dive.md & src/priority_simulation.py`

Deep Dive Track 3: Diurnal Arrivals, Dynamic Shift Staffing, and Multi-Department Network Allocation

Expands Questions 8, 9, and 10 & Flagship Research Theme 1

Stochastic Framework: Non-Homogeneous Poisson Process (NHPP), Pointwise Stationary Approximation (PSA) across 3 shifts, 4-Station Open Jackson Queueing Network, and Fox's Separable Convex Resource Optimization.

Empirical Datasets: Calibrated diurnal arrival profile from NYS SPARCS / MIMIC-ED ($■ar{\lambda}=2.083$, $A=1.167$, peak at 14:12); 4-station hospital network transfer parameters; Multi-department offered loads ($a = [20, 30, 150]$ Erlangs, clinical weights $w = [4, 2, 1]$).

Algorithms & Methods: Lewis–Shedler acceptance–rejection thinning; PSA hourly Erlang-C evaluation; Linear traffic equation solver $(I - P^T)\vec{\Lambda} = \vec{\gamma}$ and spectral stability audit; Fox's Greedy Marginal Allocation algorithm.

Key Discoveries & Audited Findings:
1. **Dynamic 3-Shift Staffing:** Shifting 25 beds from quiet night hours to day (14 night / 22 day / 18 evening) cuts peak wait from **41.5 min down to 1.1 min (-97.3%)** under **identical 432 daily bed-hours** ($0 budget increase).
2. **Jackson Network Traffic & Instability Discovery:** Exact linear solution is $\vec{\Lambda} = [40.0, 6.229, 8.538, 30.126]$ pts/day. Nominal MedSurg capacity ($c=120$) is **unstable ($\rho_4 = 125.52\% > 1.0$)**, proving that ≥ 151 beds are required for basic stability and ≥ 178 beds for 85% occupancy.
3. **Fox Bed Allocation ($C = 230$ beds):** Optimal allocation $\vec{c} = [33, 43, 154]$ reduces weighted hospital blocking from 0.4037 (proportional) to 0.0655—an **83.8% loss reduction**.

Source Document: `deep_dives/03_diurnal_staffing_and_network_deep_dive.md & deep_dives/verify_deep_dives.py`